

BishalAnalytics2005 6 X

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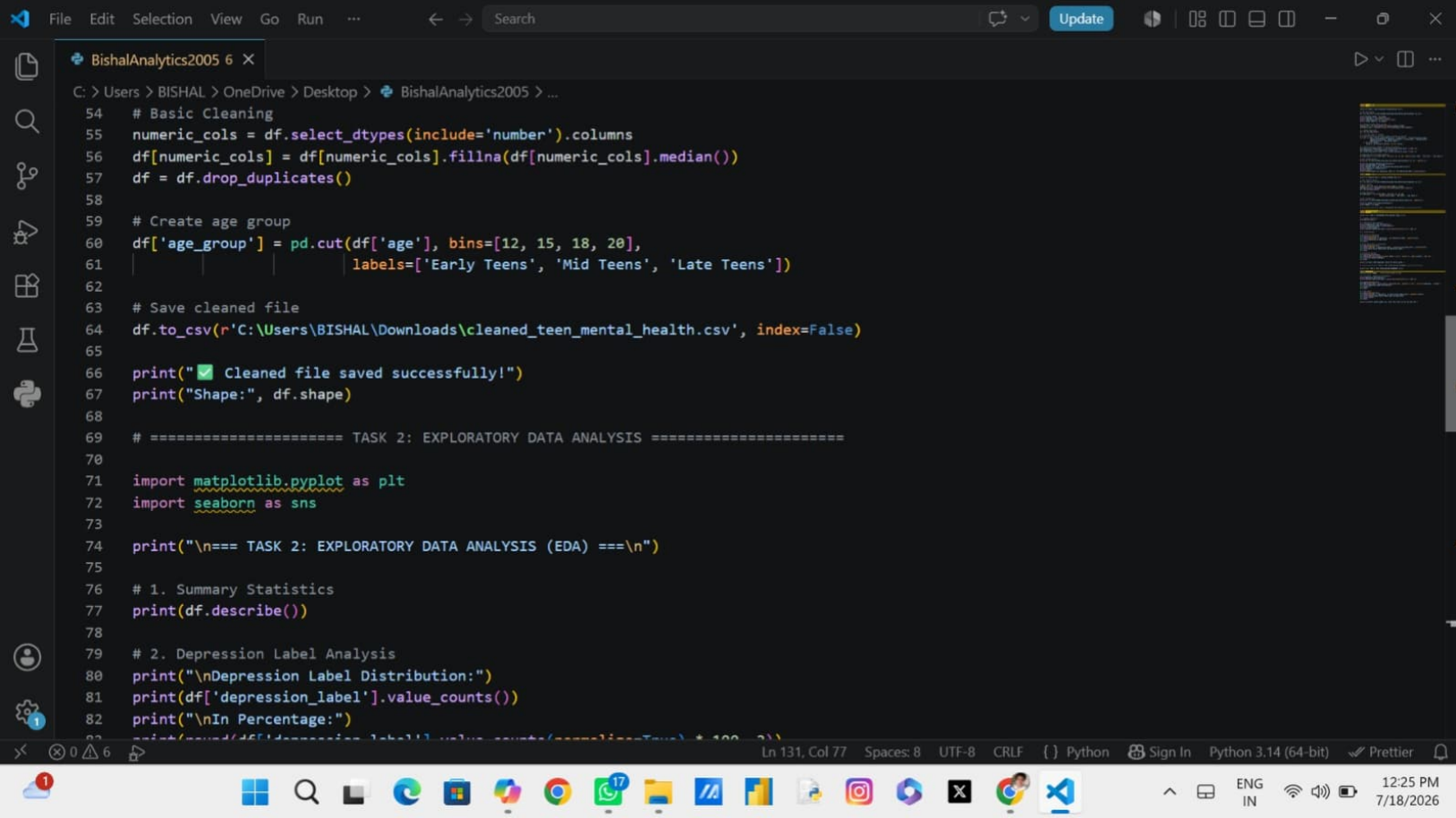
```
1 import pandas as pd
2 import numpy as np
3
4 print("=== TASK 1: Data Cleaning & Preprocessing ===\n")
5
6 # Load your dataset
7 df = pd.read_csv('C:\Users\BISHAL\Downloads\Teen_Mental_Health_Dataset (3).csv')
8
9 print("Original Shape:", df.shape)
10 print("\nColumns:", df.columns.tolist())
11 print("\nMissing Values:\n", df.isnull().sum())
12 print("\nData Types:\n", df.dtypes)
13
14 # 1. Handle missing values (if any)
15 numeric_cols = df.select_dtypes(include=np.number).columns
16 df[numeric_cols] = df[numeric_cols].fillna(df[numeric_cols].median())
17
18 # 2. Remove duplicates
19 df = df.drop_duplicates()
20
21 # 3. Fix data types (if needed)
22 # Already mostly good, but ensure numeric columns are correct
23 for col in ['age', 'daily_social_media_hours', 'sleep_hours', 'screen_time_before_sleep',
24            'academic_performance', 'physical_activity', 'stress_level', 'anxiety_level',
25            'addiction_level', 'depression_label']:
26     if col in df.columns:
27         df[col] = pd.to_numeric(df[col], errors='coerce')
28
29 # 4. Handle outliers (example for hours and scores)
```

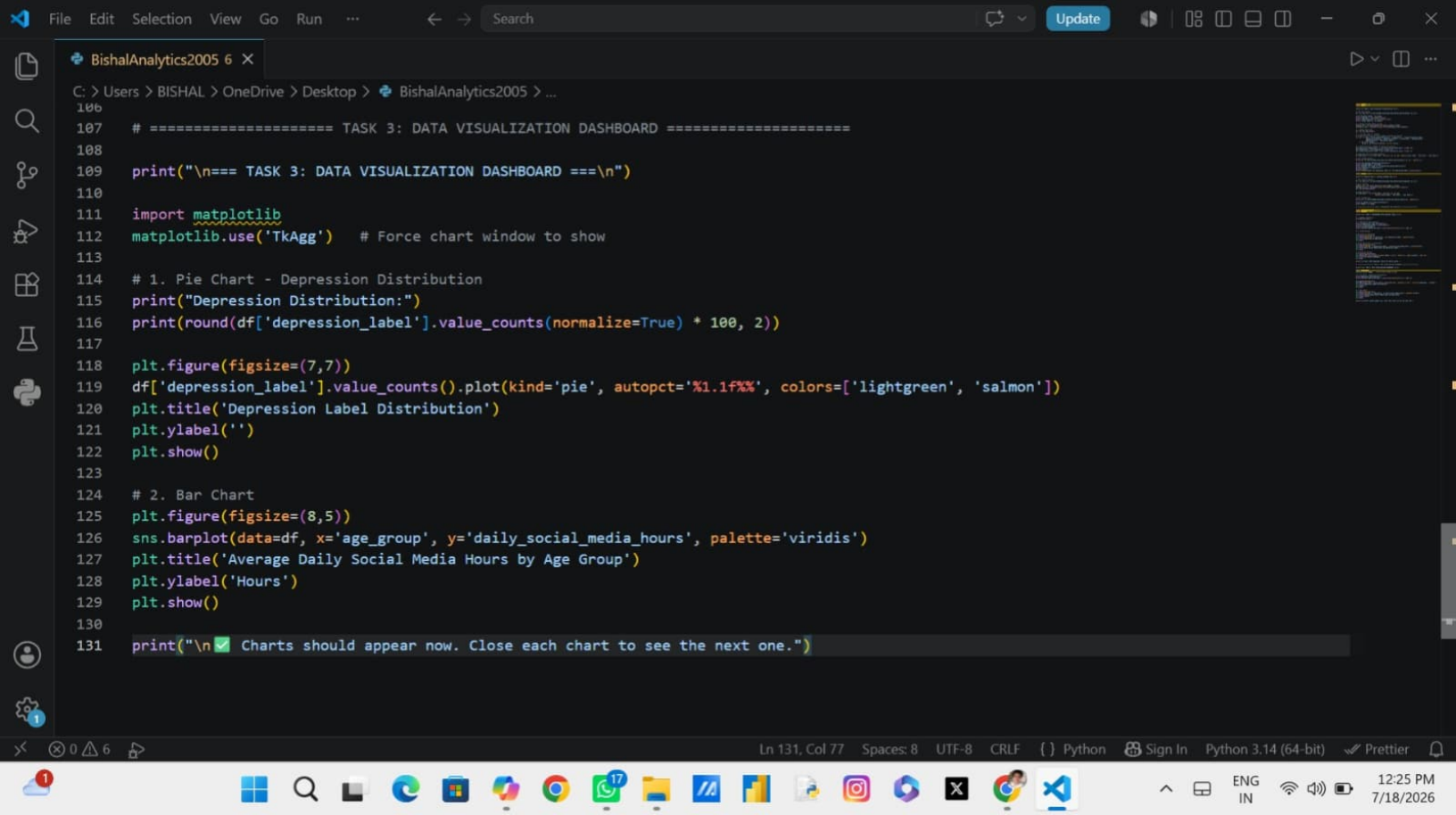
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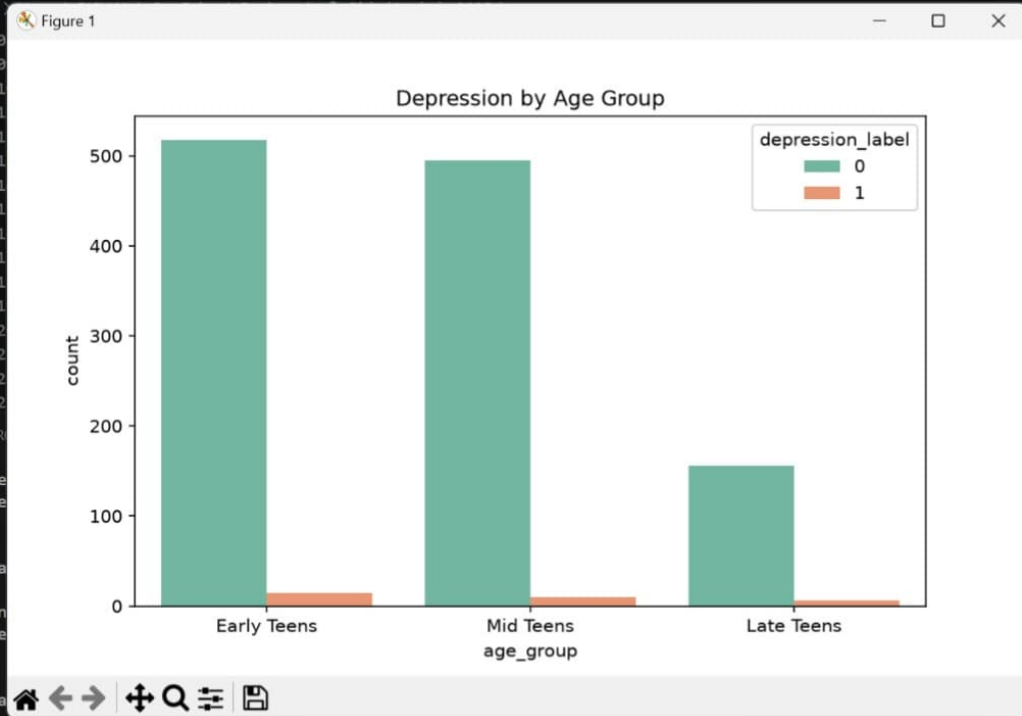
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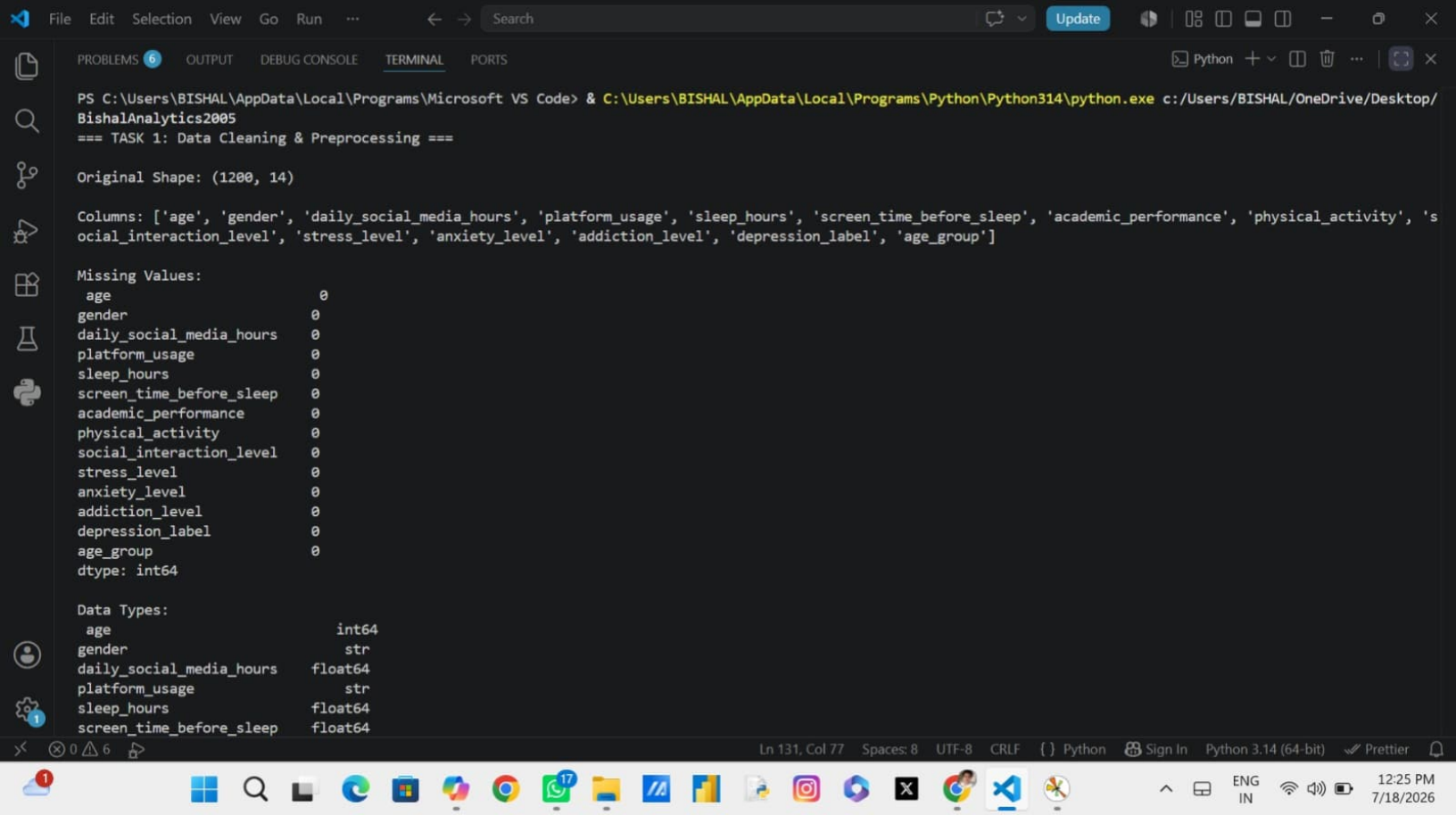
```
29 # 4. Handle outliers (example for hours and scores)
30 df['daily_social_media_hours'] = df['daily_social_media_hours'].clip(0, 12)
31 df['sleep_hours'] = df['sleep_hours'].clip(3, 12)
32 df['screen_time_before_sleep'] = df['screen_time_before_sleep'].clip(0, 5)
33
34 # Create age group for better analysis
35 df['age_group'] = pd.cut(df['age'], bins=[12, 15, 18, 20], labels=['Early Teens', 'Mid Teens', 'Late Teens'])
36
37 # Save cleaned dataset
38 df.to_csv(r'C:\Users\BISHAL\Downloads\Teen_Mental_Health_Dataset (3).csv', index=False)
39
40 print("\n✅ Cleaning Completed Successfully!")
41 print("Cleaned Shape:", df.shape)
42 print("Cleaned file saved as: **cleaned_teen_mental_health.csv**")
43 print("\nPreview of Cleaned Data:")
44 print(df.head())
45 print("\nValue Counts for Depression Label:\n", df['depression_label'].value_counts())
46
47 import pandas as pd
48
49 print("=== Running Task 1: Creating Cleaned File ===")
50
51 # Load original dataset
52 df = pd.read_csv(r'C:\Users\BISHAL\Downloads\Teen_Mental_Health_Dataset (3).csv')
53
54 # Basic Cleaning
55 numeric_cols = df.select_dtypes(include='number').columns
56 df[numeric_cols] = df[numeric_cols].fillna(df[numeric_cols].median())
57 df = df.drop_duplicates()
```

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=== TASK 2: EXPLORATORY DATA ANALYSIS (EDA) ===

	age	daily_social_media_hours	sleep_hours	screen_time_before_sleep	...	stress_level	anxiety_level	addiction_level	depression_label
count	1200.000000	1200.000000	1200.000000	1200.000000	...	1200.000000	1200.000000	1200.000000	1200.000000
mean	15.928333	4.536667	6.449417	1.740333	...	5.445833	5.636667	5.565000	0.025833
std	2.021947	2.029599	1.442677	0.716660	...	2.903290	2.859453	2.830627	0.158704
min	13.000000	1.000000	4.000000	0.500000	...	1.000000	1.000000	1.000000	0.000000
25%	14.000000	2.800000	5.200000	1.100000	...	3.000000	3.000000	3.000000	0.000000
50%	16.000000	4.500000	6.500000	1.800000	...	5.000000	6.000000	6.000000	0.000000
75%	18.000000	6.300000	7.600000	2.400000	...	8.000000	8.000000	8.000000	0.000000
max	19.000000	8.000000	9.000000	3.000000	...	10.000000	10.000000	10.000000	1.000000

```
[8 rows x 10 columns]
```

Depression Label Distribution:

depression_label

0 1169

1 31

```
Name: count, dtype: int64
```

In Percentage:

depression_label

0	97.42
---	-------

1 2.58

```
Name: proportion, dtype: float64
```